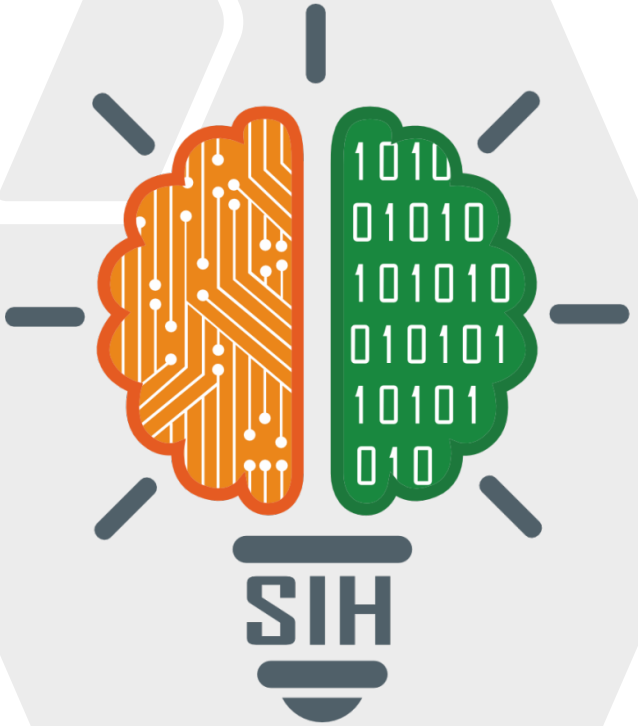
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PRIVAGENT

Privacy-first visual perception for browser agents

Problem Statement ID	SIH26171
Problem Statement Title	On-device Visual Perception for Light-weight Browser Agents
Theme	Smart Automation
PS Category	Software
Team ID	T040_BLUE_HORIZON_Software
Team Name	**BLUE HORIZON**



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PRIVAGENT

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Proposed Solution (Describe your Idea/Solution/Prototype)

01 LOCAL PERCEPTION

Read DOM and screen context on the device.

02 PRIVACY GATE

Detect and redact sensitive regions before any request.

03 HYBRID ACTION

Server plans on safe context. The client validates and acts.



Capture → DOM + screen

Inspect → Local vision

Sanitize → PII redaction

Plan → Safe context

Act → Validate locally

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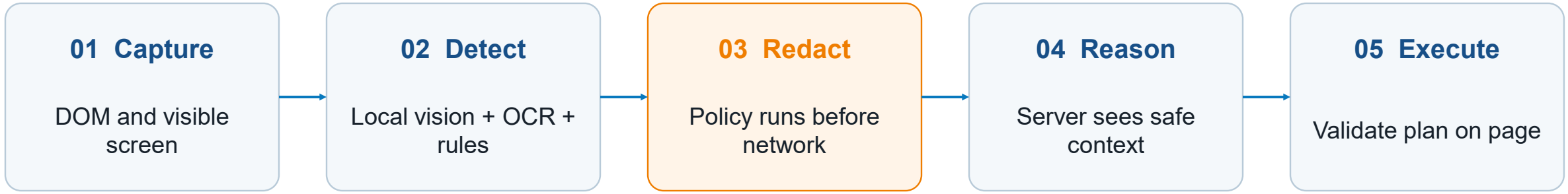
TECHNICAL APPROACH

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Technologies to be used (e.g. programming languages, frameworks, hardware)

BROWSER CLIENT	LOCAL INPUT	VISION RUNTIME	COMPUTE	SERVER
Chrome WebExtensions JavaScript	DOM + screenshot OCR + privacy rules	ONNX Runtime Web Transformers.js	WebGPU preferred WASM fallback	LLM/VLM returns constrained JSON actions

Methodology and process for implementation (Flow Charts/Images/ working prototype)



Model size, accuracy, browser support and latency will be benchmarked before a model is selected.

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FEASIBILITY AND VIABILITY

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Analysis of the feasibility of the idea

A browser extension can combine page structure with visual context.

ONNX Runtime Web supports browser inference through WebGPU and WebAssembly.

The design keeps heavy reasoning on a server while local checks protect the request.

Potential challenges and risks

PII may be missed when content appears only in a screenshot.

Model performance varies by device and browser.

A server action can be stale or unsafe for the current page.

Strategies for overcoming these challenges

Combine DOM signals, OCR, visual detection and rules.

Use a small model in a worker, with a WASM fallback.

Fail closed, then validate allow-listed actions against the live page. Block typing into protected fields.

Model choice and resource use remain benchmark decisions; the current proposal does not claim measured performance.

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IMPACT AND BENEFITS

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Potential impact on the target audience

Browser users working with private portals and dashboards.

People repeating structured web tasks.

Organizations seeking to minimize screen data sent to AI services.

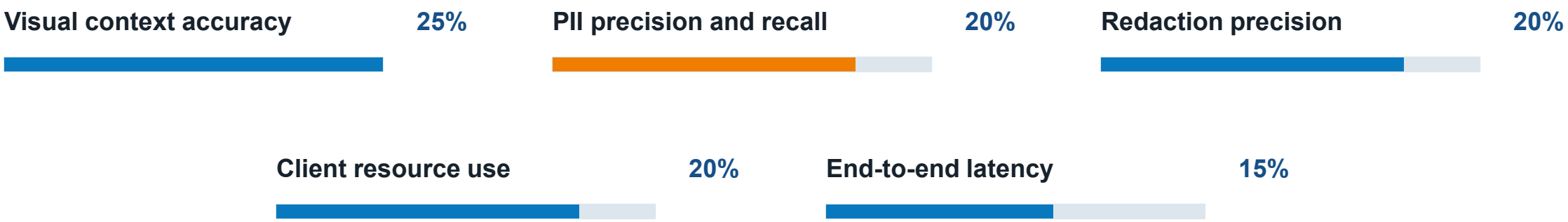
Benefits of the solution (social, economic, environmental, etc.)

Designed to keep sensitive values on-device until a privacy check clears the request.

Server reasoning uses sanitized task context.

Local policy can block unsafe actions, with an audit trail for review.

SIH evaluation weights | 100% total



Weights show judging priorities, not PRIVAGENT results. Performance targets require measured prototype data.

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RESEARCH AND REFERENCES

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Research and technical references used to shape the proposed design

SIH 2026 Problem Statements Catalogue

SIH26171 and evaluation criteria, cited in the project blueprint

ONNX Runtime Web

https://onnxruntime.ai/docs/tutorials/web/

Browser inference and WebGPU / WebAssembly execution providers

Transformers.js WebGPU guide

https://huggingface.co/docs/transformers.js/guides/webgpu

Browser model inference using WebGPU

Chrome Extensions content scripts

https://developer.chrome.com/docs/extensions/develop/concepts/content-scripts

DOM access and extension context